

MST-Technologies and Processes

August, 16th, 2021

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task
- write with a pen NOT with a pencil
- if you hinder the exam you will be removed from the lecture hall and fail the exam (5.0 with the statement "Störung")

Cheating

- any attempt of cheating results in immediate failure of the exam (5.0 with the statement "Täuschung")
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
 - in case you feel unwell during the exam, you may withdraw during the exam for medical reasons; you need to provide (within 3 days) a written note from your doctor confirming the withdrawal for medical reasons
 - please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
 - you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>
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Necessary Points: **55 of 110 in total**

Points: _____

Result: _____

1. Processes

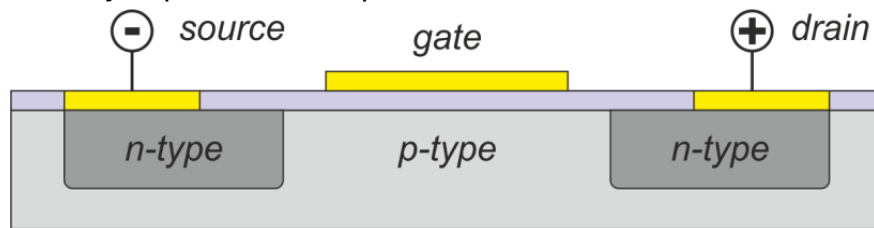
28



1.1 Manufacturing of a field effect transistor (FET) 22

Sketch the manufacturing steps required to fabricate the following structure which is a MOSFET. It is manufactured on a silicon substrate, the gates are made from gold. Please note that we discussed a very short variant of the manufacturing process in the lecture - the full manufacturing workflow is longer. Each manufacturing process (e.g., lithography) used should be depicted out entirely when you first use it. If you use the same process again, you do not need to redraw the entire process.

Sketch and briefly explain each step.



1.2 We have three different categories of processes in MST, name them and sort the following processes into the right category: 6

- Metallization
- Development during lithography
- Thermal oxidation



2. Thermal oxidation of silicon

16

2.1 In the Deal-Grove model, the following variables are used:

4

- D as the diffusivity of the oxidant in silicon dioxide
- k as the reaction rate of the oxidation of silicon to silicon dioxide
- N as the number of oxidant molecules required to produce one unit volume of oxide
- C as the ambient concentration of the oxidant
- $A = \frac{2D}{k}$ as a simplification factor
- $B = \frac{2DC}{N}$ as a second simplification factor

State the equation for the layer thickness $x(t)$ at lower layer thicknesses, i.e., for $x(t < t_c)$.

2.2 For usage in the Deal-Grove model we determined the following factors:

4

- $A = 0.5 \mu\text{m}$
- $B = 0.25 \mu\text{m}^2/\text{h}$

What is the characteristic time t_c ? *[You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]*

2.3 Using the same parameters as above, we now run an oxidation process for 15 minutes (which is shorter than t_c). What is the expected layer thickness we achieve? *[You do not need a calculator for this - if you find odd numbers, just write down the equation with the numbers]*

4

2.3 Sketch and explain the setup for dry oxidation of silicon. Make sure to include all relevant labels in your sketch.

4



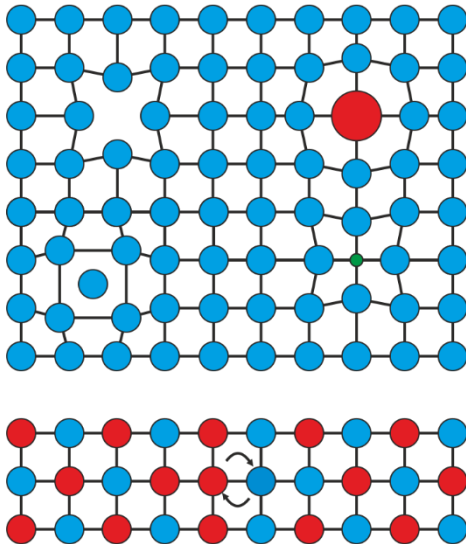
3. Semiconductor materials

30

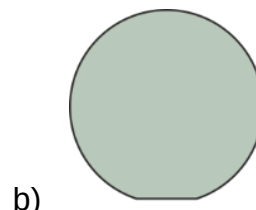
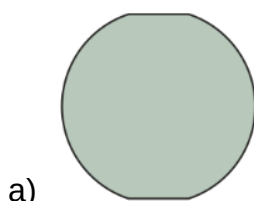
- 3.1 How many binding partners does silicon have, i.e., what is the number of valence electrons? 6
What is the most common form of hybridization for silicon?
What is the crystal structure of silicon?

- 3.2 The image shows zero-dimensional defects. 7

- what is the defect dimension?
- label the five defect types?



- 3.3 Indicate the following wafers by their wafer flats 4



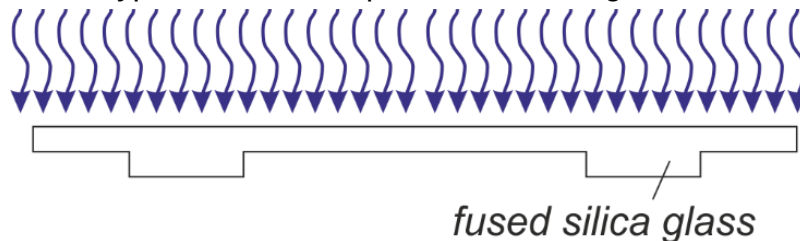
- 3.4 Silicon-On-Insulator wafers have a wide use case in semiconductor manufacturing. Why and what are they being used? 4
- 3.5 Sketch and explain the SIMOX process. Make sure to include all relevant labels in your sketch as well as an explanation of each step. 9



4. Photolithography 28

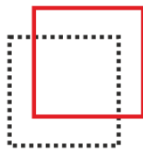
- 4.1 Sketch the setup for a projection lithography system. Make sure to label each component in your sketch. 6
- 4.2 Sketch the typical photolithography workflow for the generation of a metal etching process. Make sure to label each component in your sketch and include a short explanation to each step. 10

- 4.3 Which type of mask is depicted in this image? 1



- 4.4 What are the two parameters that a mask can modulate on the incoming light? 2
- 4.5 If projecting a reticle as part of a photolithography using a stepper, we often come across several potential errors. Label the following errors (1 point each) and indicate what the underlying problem is (2 points). 9

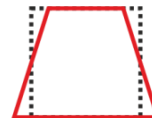
a)



b)



c)



5. Cleanroom Technology 8

- 5.1 Name four general recommendations for working in the cleanroom 4
- 5.2 Sketch and explain the setup of a Filter Fan Unit (FFU). Make sure to label each component in your sketch and include a short explanation. 4

